

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (CL)

May 2018

CL 316 Highway Engineering

Date: 01.05.2018, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Choose the correct option from the following multiple choice questions: [07]

(1) A research organisation formed during the time of independence for the research and development works related to highway was _____.

- a) Indian Road Congress b) Central Road Research Institute
c) Highway Research Board d) National Highway Authority of India

(2) Which is the most flexible type of transportation available?

- a) Roadway b) Railway c) Waterway d) Airway

(3) Which type of roads can be utilized during monsoon?

- a) Unpaved roads b) Gravel roads c) Fair weather roads d) Bituminous roads

(4) The arterial roads are classified as _____.

- a) Rural roads b) Urban Roads c) National highway d) State highway

(5) Road density of second 20-years road plan is _____.

- a) 32km per 100 km² b) 42km per 100km² c) 52km per 100 km² d) 22km per 100km²

(6) The economical option during the construction of a road around a hill is _____.

- a) Cut the hill b) Provide a tunnel
c) Provide a road around the hill d) Look for other alternative approach

(7) The sequence of four stages of survey in a highway alignment is _____.

- a) reconnaissance, map study, preliminary survey and detailed survey
b) map study, preliminary survey, reconnaissance and detailed survey
c) map study, reconnaissance, preliminary survey and detailed survey
d) preliminary survey, map study, reconnaissance and detailed survey

Q - 2 (a) List out the various steps in sequential order for the conventional approach method of [02]

preliminary survey for finalizing the highway alignment.

- (b) A State Highway passing through a rolling terrain has a horizontal curve of radius equal to the ruling minimum radius and design speed of 80 kmph. [08]

(i) Design all the geometric features of this curve, assuming suitable details.

(ii) Specify minimum set-back distance from the center line of the two lane highway on the inner side of the curve up to which the buildings etc. obstructing vision should not be constructed so that intermediate sight distance is available throughout the circular curve. Assume the length of circular curve greater than the sight distance

OR

- (b) Derive an expression for calculating overtaking sight distance on a highway. Why are overtaking zones provided? What is the basis of deciding its length? Also draw a neat sketch and show the signs to be installed and their positions. [08]

- (c) Illustrate the neat sketch of Cross-Section of road in cutting showing all its components. [04]

- Q - 3 (a) What are the requirements of a good highway drainage system? [04]

- (b) **Mention the causes and suggest the remedial measures of following failures in flexible pavements:** [05]

(1) Alligator Cracking (2) Reflection Cracking (3) Rutting
(4) Waves and Corrugation (5) Raveling of Aggregate

- (c) Discuss the desirable properties of aggregates used in highway construction. [05]

OR

- Q - 3 (a) Define Surface highway drainage and Sub-surface highway drainage. Discuss the importance of highway drainage. [04]

- (b) Why maintenance is required in highway pavements? Enlist and explain the types of highway maintenance. [05]

- (c) Briefly explain the principle of the various tests on road stones. Also specify desirable values of test results. [05]

SECTION -II

- Q - 4 **Choose the correct option from the following multiple choice questions:** [07]

(1) The traffic volume is usually expressed in _____ .

a) LMV b) PCU c) LCV d) HCV

(2) The first objective of the traffic volume studies is _____ .

a) To decide priority for improvement of roads b) For geometric design
c) For computing roadway capacity d) To plan traffic operation

- (3) The traffic flow is _____.
 a) Static b) Dynamic c) Static and dynamic d) May be static or dynamic
- (4) No parking sign is a sign of _____.
 a) Warning Sign b) Regulatory Sign c) Informatory Sign d) Indication Sign
- (5) Traffic lane is the portion of the road over which _____ traffic move.
 a) Vehicular b) Pedestrian c) Animal d) None of the above
- (6) Number of vehicles passing from particular point of the road section is called _____.
 a) Traffic volume b) Traffic density c) Traffic capacity d) Traffic flow
- (7) What is the angle for acuity of vision laid down by the licensing authority on highways?
 a) 1° b) 2° c) 3° d) 4°

- Q - 5 (a)** Discuss the importance of highway economy studies. [05]
- (b)** Enlist the layers of rigid pavement with neat sketch. Also write down the functions of each layer. [04]
- (c) Define:** [05]
- (1) Prime Coat (2) Tack Coat (3) Seal Coat
- (4) CBR at 2.5mm penetration (5) Modulus of subgrade reaction

OR

- Q - 5 (a)** Write a note on Highway Financing. [05]
- (b)** What are the various factors to be considered in pavement design? Discuss significance of each factor. [04]
- (c)** Specify the materials required for the construction of WBM Road. Enumerate the steps for construction of WBM Road. [05]

- Q - 6 Answer the following questions (Any two):** [14]

- (a)** Explain the relationship between speed, travel time, volume, density and capacity with neat sketch.
- (b)** Explain the concept of level of service while deciding the design capacity of a road.
- (c)** Briefly outline the IRC recommendations for determining the thickness of cement concrete pavements.
- (d)** Compare flexible pavement and rigid pavement.
